

[4457] - 212

S.E. (Computer Engg.) (Semester - I)

DATA STRUCTURES AND PROBLEM SOLVING

(2012 Course)

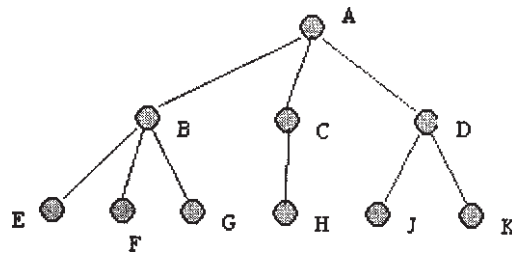
Time : 2 Hours]

[Max. Marks : 50

Instructions to the candidates:-

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Draw the IPO chart for obtaining the solution for “Calculating the area of Triangle’. [3]
- b) Define the terms with respect to sorting: internal sort, external sort and sorting stability. [3]
- c) Convert the following generalized tree into a binary tree. [3]



- d) What is the necessity of threaded binary tree? Define the terms inorder successor and inorder predecessor with respect to inorder threaded binary tree. [3]

OR

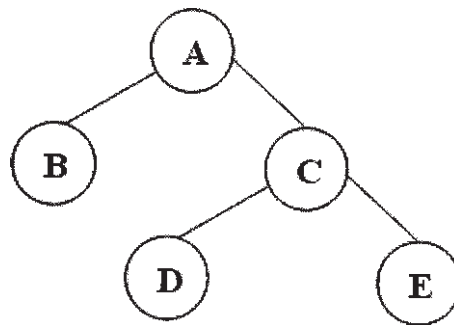
Q2) a) Find the frequency count of the following code : **[3]**

```
for(i=0;i<n;i++)
{
    for(j=0;j<n;j++)
    {
        c[i][j] = 0;
        for(k=0;k<n;k++)
        {
            c[i][j] = a[i][k] * b[k][j];
        }
    }
}
```

b) Sort the following data in ascending order using Radix Sort:

25, 06, 45, 60, 140, 50, **[3]**

c) Represent the following binary tree using array. **[3]**



d) Write a pseudo C/C++ code for any of the recursive depth first traversal of the binary tree. **[3]**

Q3) a) Explain any three applications of graphs in the area of Computer Engineering. **[3]**

b) Differentiate between Prim's and Kruskal's algorithm for generating the spanning tree of the graph. **[3]**

c) Create an AVL tree for the following data by inserting it in an order one at a time :

10, 20, 15, 12, 25, 30. **[4]**

d) Enlist the names of static tree tables with suitable example. **[2]**

OR

- Q4)** a) Explain the situation in which linked representation of a graph is more beneficial than array representation. [2]
- b) Write a pseudo C/C++ code for finding depth first traversal of the graph. [4]
- c) What is the use of hash tables? Enlist the characteristics of good hash function. [3]
- d) Assume the size of hash table as 8. The hash function to be used to calculate the hash value of the data X is: $X\%8$. Insert the following values in hash table: 10, 12, 20, 18, 15. Use linear probing without replacement for handling collision. [3]

- Q5)** a) Write a pseudo C/C++ code to sort the data in ascending order using heap sort. [6]
- b) Create a B tree of order 3 for the following data : 20, 10, 30, 15, 12, 40, 50. [4]
- c) Explain the various modes of opening the file in C or C++. [3]

OR

- Q6)** a) Sort the following data in ascending order using heap sort: 15, 10, 40, 25. [4]
- b) Write a pseudo C/C++ code to search the data stored in a B tree. [6]
- c) Explain any three operations carried out on sequential files. [3]

- Q7)** a) Write a parallel algorithm to calculate the addition of numbers stored in an array using prefix computation method. [6]
- b) Explain the various models used for parallel computation. [4]
- c) Explain pointer doubling problem in brief. [3]

OR

- Q8)** a) Write a parallel algorithm for odd - even merge sort. Explain the algorithm with suitable Example. [7]
- b) Write a parallel algorithm to perform the addition of the given numbers using complete binary tree method. [6]

